The Best Queues

Cont-de Larrard: direction, timing and volatility from two queues

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The question

In a liquid large-tick stock almost all activity happens at the best bid and ask. **Can two queues explain the next price move, its timing, and volatility?**

- The direction of the next move is what a quote should anticipate.
- The time to the next move sets how long a quote is exposed.
- The volatility formula links microstructure (rates, depth) to the risk an LP carries over minutes.

In this deck

1. First passage of a queue to zero.
2. $p_{\uparrow}$, durations, diffusion limit.
3. Fitting it to a richer book, and the statistics of conditional frequencies.
4. Evidence from Cont and de Larrard (2013).

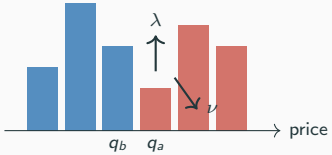
The model

One-tick spread. The bid queue q_b and ask queue q_a (orders) each move independently:

$$q \to q+1 \text{ at rate } \lambda, \qquad q \to q-1 \text{ at rate } \nu = \mu + \theta,$$

with λ the limit order rate per queue, μ the market order rate and θ the cancellation rate, all in s^{-1} and *constant* (not per order).

- When the ask empties the price moves up one tick δ; when the bid empties it moves down.
- After a move both queues are redrawn from a distribution f on $\mathbb{N}^2$; this "regeneration" is what makes the price a random walk.



First passage and the next move

A queue of size q is a birth-death walk; its time τ_q (s) to hit zero has density

$$f_q(t) = \frac{q}{t}\left(\frac{\nu}{\lambda}\right)^{q/2} I_q\big(2\sqrt{\lambda\nu}\,t\big)\, e^{-(\lambda+\nu)t},$$

with I_q the modified Bessel function of the first kind. The queues are independent, so

$$p_{\uparrow}(q_b, q_a) = \int_0^{\infty} f_{q_a}(t)\, P(\tau_{q_b} > t)\, \mathrm{d}t,$$

$$P(\tau > t \mid q_b, q_a) = P(\tau_{q_b} > t)\, P(\tau_{q_a} > t).$$

Balanced case $\lambda = \nu$, large queues: (q_b, q_a) is close to a planar Brownian motion stopped on the axes, and

$$p_{\uparrow} \approx \tfrac{2}{\pi} \arctan(q_b/q_a).$$

Heavy tail. $P(\tau > t) \approx q_b q_a/(\pi\lambda t)$: the mean time between moves is infinite.

Unbalanced $\lambda < \nu$: queues drain on average, $\mathbb{E}[\tau_q] = q/(\nu - \lambda)$, and durations have a light tail.

Volatility from order flow

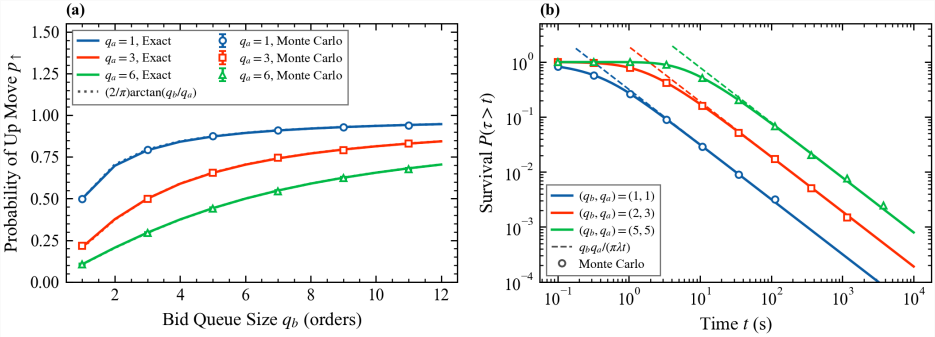
Balanced case. Let $M_F = \sum_{x,y} x\, y\, f(x,y)$ (orders^2) be the mean product of queue sizes after a move. Because $P(\tau > t) \sim M_F/(\pi\lambda t)$, the truncated mean duration grows like $(M_F/\pi\lambda)\ln(\lambda t)$, and the number of moves by time t like $\pi\lambda t/(M_F \ln(\lambda t))$. Each move is $\pm\delta$, so for independent moves over a horizon T (s)

$$\mathrm{Var}(S_T - S_0) \approx \frac{\pi\lambda\,\delta^2}{M_F}\,\frac{T}{\ln(\lambda T)} \quad\Longrightarrow\quad \sigma^2 \propto \frac{\lambda\,\delta^2}{M_F}.$$

- Cont and de Larrard (2013) prove the diffusion limit on the time scale $t\, n \ln n$.
- Reading: more limit order flow raises volatility; deeper queues after a move lower it.
- If increments are negatively correlated (bid queue usually smaller than the new ask queue after an up move), the walk mean-reverts at short lags.

Not checked numerically here: the $\ln T$ correction makes the diffusion limit slow to reach in simulation.

Closed forms against simulation



- Monte Carlo (20,000 paths per point) matches the closed form within 2.4 standard errors; an exact sparse linear solve agrees to 2×10^{-3}.
- The duration tail $q_b q_a/(\pi\lambda t)$ is within 0.88% of the exact survival at 10^4 s. The arctangent law is off by 0.0017 at $(2, 5)$.

Statistics: fitting the model, and samples that are not independent

Fitting. From one-tick-spread periods of total length T_1 (s):

$$\hat{\lambda} = \frac{\#\{\text{orders joining a best queue}\}}{2T_1},$$

$$\hat{\nu} = \frac{\#\{\text{market orders} + \text{cancels at the best}\}}{2T_1}.$$

Then $\hat{p}_\uparrow(q_b, q_a)$ from the closed form.

Testing. Compare with the empirical frequency of up moves given (q_b, q_a).

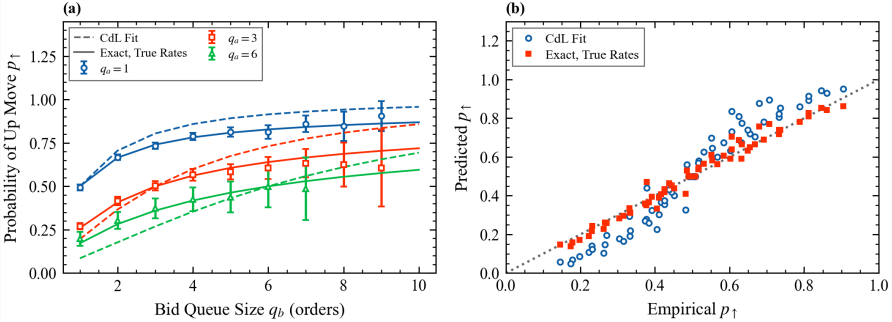
The trap. If every book event is a sample, all samples between two price moves share one outcome. In the simulated book, state $(2, 5)$ has thousands of samples but only about a minute of exposure.

Binomial errors $\sqrt{p(1-p)/n}$ are then far too small. Treat all samples before the same move as one **cluster**:

$$\widehat{\mathrm{Var}}(\hat{p}) = \frac{1}{n^2} \sum_c \Big(\sum_{i \in c} (y_i - \hat{p}) \Big)^2,$$

with $y_i \in \{0, 1\}$ the outcome of sample i.

Fitting constant rates to a book whose rates are not constant



Power-law Markovian book (deck 1), 1,500,000 events, 34,204 mid moves. In it the best queues shrink at $\mu + \theta q$, which grows with q. Fitted CdL rates: $\hat{\lambda} = 1.903$, $\hat{\nu} = 2.302$ s^{-1}.

Over 64 cells, CdL is off by 0.089 on average (worst 0.23); the exact answer for state-dependent rates is off by 0.023, within 2.1 clustered s.e. CdL overstates how decisive imbalance is: with ν constant, long queues drain too slowly.

What the data show

Cont and de Larrard (2013), NYSE stocks on 26 June 2008:

- **One-tick spread most of the time**: 98.82% of observations for Citigroup, 98.80% for General Electric, 98.71% for General Motors.
- **Nearly balanced flow**: the estimated limit order rate is slightly below the rate of market orders plus cancellations, with a small gap.
- **Asymmetric regeneration**: after an up move the new bid queue is usually smaller than the new ask queue (for Citigroup the relevant probability exceeds 0.7), so price increments are negatively correlated at high frequency.
- **Volatility across stocks**: for Dow Jones stocks the standard deviation of 10-minute price changes rises roughly in proportion to the rate-over-depth ratio predicted by the model.
- Conditional probabilities of an up move given the queues agree well with Citigroup tick data.

The arctangent shape is also the basis of the **micro-price** (Stoikov, 2018), deck 6.

Discussion

What it gets right

- Imbalance predicts direction, with a closed form.
- Heavy-tailed durations: long quiet spells between moves.
- A volatility formula with no price data in it.

What it leaves out

- Depletion that depends on queue size (fixed by the exact state-dependent solve here, or by queue-reactive models, deck 3).
- Memory and clustering of events (deck 4).
- Spreads wider than one tick, so small-tick assets are out of scope.

For a liquidity provider

- $p_\uparrow$ is the probability that a resting bid becomes stale; use the state-dependent version when cancellation scales with queue size.
- The duration law sets how long a quote sits exposed; a $1/t$ tail means occasional very long exposures.
- Conditional frequencies from tick data need clustered errors, or every cell looks significant.

Open question. A tractable version with history (Hawkes) and state dependence that keeps closed forms for $p_\uparrow$.

Reproduce

Code: `best_queues.py` (closed forms, simulation), `queue_position.p_up_state_dependent` (exact solve for $\mu + \theta q$), `markov_lob.empirical_p_up` (frequencies with clustered errors). Experiments: `e2_best_queues.py`, `e23_cdl_on_book.py`.

```
from stochastic_lob import best_queues as bq
bq.p_up(q_b=2, q_a=5, lam=1.0, nu=1.2)      # P(next move is up)
bq.duration_survival(t=60.0, q_b=2, q_a=5, lam=1.0, nu=1.0)
```

References i

Rama Cont and Adrien de Larrard. Price dynamics in a Markovian limit order market. *SIAM Journal on Financial Mathematics*, 4(1):1–25, 2013.

Sasha Stoikov. The micro-price: a high-frequency estimator of future prices. *Quantitative Finance*, 18(12):1959–1966, 2018.